

A Smart and Private Blockchain-enabled Framework for Digital Assets

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ABSTRACT

Nowadays, many digital assets are generated all the time, and managing these digital assets has become an important issue. The traditional digital asset management system manages data by the service provider, and data is not processed transparently. Furthermore, centralized storage makes Single Point of Failure (SPOF), and it is also challenging to ensure that data backed up by all nodes are synchronized. Rapid improvement of blockchain technology, the above two problems are resolved by leveraging blockchain's decentralization and tamper resistance characteristics. The EU General Data Protection Rules (GDPR) state the principles for protecting data owners in 2018. Several principles for deleting personal data conflict with the blockchain, such as Storage Limit and the Right to Be Forgotten. The existing blockchain-enabled digital asset management methods do not apply to the GDPR. Although some methods are proposed for storing digital assets in off-chain storage and storing the hash address of storage in the blockchain, the problem of SPOF has not been resolved yet. This motivates to propose SPChain, a smarter and private GDPR-compliant digital asset framework enabled by blockchain and decentralized InterPlanetary File System is implemented to solve the problem of SPOF. The proposed SPChain to digital art management provides an implementation using Hyperledger Fabric. The proposed framework consists of a model developer, art creator, collector, service provider and securely assessed AI model for managing digital assets. The experiments measured throughput, latency and resource consumption of various functions in the smart contract's artwork_CC, walle_CC, 3A_CC, log_CC, model_CC. This proposed work is the first novel work to combine digital asset management with detailed design of the decentralized off-chain storage and artificial intelligence models.

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CCS CONCEPTS

• **Blockchain** ; • **GDPR**; • **Digital Asset management**;

KEYWORDS

Blockchain, Private Blockchain, AI Smart Contracts

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1 INTRODUCTION

Numerous items have been digitized around our lives. The digital assets (DS) are managed by Service Providers (SPs) or digital asset management systems (DAMS). Digital asset management is used to access and manage queries in an organized way quickly. In the digital assessment system, two issues are raised, such as DS not being transparent and data services in the centralized, single point of failure (SPOF). In May 2018, the General Data Protection Regulation (GDPR), whose effect area covered EU residents to provide services to all European countries. Blockchain technology has recently become the rage. Due to the tamper-resistant design of the ledger maintained by the nodes in a blockchain network, all changes or transactions are detected. Different works are proposed to resolve the blockchain and GDPR conflict issues. The main background of this proposed work is Blockchain technology. Due to the tamper-resistant design of the ledger maintained by the blockchain network, all the transactions and changes are detected [23]. The dominant application of blockchain is in the financial sector, healthcare, and cryptocurrencies [18]. Comparing this digital with original digital art, "rare" digital art (also known as crypto art) makes the artwork collectable in limited quantities by attaching a unique token generated by blockchain [7]. This combination ensures the provenance of artwork and cannot be tampered with the elimination of SPOF due to the decentralization and transparency of the blockchain. The existing crypto art gallery platform such as Cryptograffiti [9] and SuperRare [15]. Both platforms have drawn art to collector and creator attention successfully integrated with Ethereum well, and keep functioning of SCs such as uploading, transferring, and updating digital assets, which restricts the art creators from generating more

artwork applications. The SC topology of Ethereum [13] contracts are, in fact, direct copies of the others. It provides better design and compliance with GDPR is the need of the hour. The three layers of blockchains with different functionaries provide the Right to Portability and the Right to be Forgotten [6]. The off-chain technique achieves GDPR compliance cooperation using SCs [2],[20]. In fact, the central idea of blockchain — decentralization, gives the same concept of returning data control to DSs in the GDPR. Nevertheless, the process of their realization conflicts with each other (e.g., the Right to be Forgotten versus tamper-proofing). In particular, the European Parliament Research Service (EPRS) has proposed blockchain-related research. In contrast to traditional SCs, AI SCs are combined with AI models to increase creativity and orient to diverse applications through invoking model inference API inside the contracts. In particular, using neural networks such as Convolutional Neural Networks (CNNs) [11] or Generative Adversarial Networks (GANs) [10] to transfer the style of artworks or even create new artifacts. SPChain also integrates the thought of exchanging machine learning model online with digital art management platforms. Details of current marketplaces for exchanging and selling AI modes are discussed in [14]. Most marketplaces are integrated with the virtualized container technology since the advantages of virtualization not only ensure programs and services output the same results in heterogeneous environments but are more lightweight than traditional virtual machines.

The rest of the paper is organized as follows. Section 2 introduces related works on digital art about blockchain, AI-based artworks, DAM with blockchain, blockchain for privacy-preserving and GDPR. Section 3 details the proposed SPChain framework, and Section 4 provides the results, discussions, and performance evaluations, followed by the conclusion and future work in Section 5.

2 RELATED WORK

The study focuses on Arts and Blockchain, AI-based Artworks, DAM with Blockchain, and Blockchain for Privacy and GDPR. The use of Blockchain for privacy-preserving and GDPR will be reviewed. Since Cryptocurrencies have become popular due to the characteristics of decentralization and security, blockchain technology is widely associated with the arts. For instance, Cryptograffiti [9] first appended QR codes to street and fine artworks. Passengers could Scanning the codes, then paying with Bitcoin as a donation. This type of crowdfunding would democratize the art market, which has thus far only been open to collectors who have the techniques and contacts for finding work after graduation, increasingly blockchain-related as a result, rare digital art emerged, as depicted in Arts meets Blockchain. With the fast growth of computer vision, it is now possible to combine art and AI. Many creators are using AI to automatically generate more realistic content. artifacts or abstract artifacts, namely AI-based artworks. Elgammal et al.[5]provide an example. GANs are proposed as Creative Adversarial Networks (CANs) based on GANs. The reality of those Artifacts cannot even be distinguished by contemporary artists. Detailed evolution Such combinations are elaborated in AI-based artworks.

2.1 Arts meets Blockchain

According to [17], there are two different ways to introduce blockchain into Arts which are: (1) Art about blockchain (2) Blockchain-based art. Wang et al. [21] presented ArtChain, which is the first blockchain-based art trading platform in Australia with a complete network incentive mechanism (i.e., POI) and privacy control design. The platform was built on Ethereum private blockchain and chose Proof of Authority as its consensus algorithm. Based on their API test, 116 times of function post_new_artworks() took approximately 18 seconds and 58 times of function buy_tokens() took about 5 seconds per node. The bottleneck resulted from highly used memory and the performance of Level-DB used in Etherum as data increased. Compare to the physical art or fine art discussed above, crypto art is entirely connected with specific and unique tokens on blockchain where tokens represent provenance for a piece of digital art [7]. Typically, such combination will be linked to another distribute system named InterPlanetary File System (IPFS) [3].

2.2 AI-based Artworks

Since AlexNet and ResNet has come out on top at the image classification tasks held by ImageNet project in 2012 and 2015 separately, the number of research on deep learning increases significantly. Radford et al. [16] proposed Deep Convolutional Generative Adversarial Networks (DCGANs) with unsupervised learning that leaved out the labels of images. Technically, DCGANs replaced any pooling layers with strided and fractional-strided convolutions. Also, batch normalization and moderate activation function in generator and discriminator brought about the stability when training for deeper generative models. At a later time, CANs [5] that based on the network architecture of DCGANs was presented. CANs redefined an entropy loss that results very large loss if generated images were classified into any high confidence style classes. In this way the artifacts became more creative but were closely bounded with enough recognition of art.

Aside from synthesis and generating artifacts, many works also do search on style transfer with machine learning. Gatys et al. [8] introduced A Neural Algorithm of Artistic Style (N.A.A.S), which converted the artistic style of images and preserved detailed contexts successfully at the same time. To elaborate, N.A.A.S combined the content of one image with the style of another by jointly minimizing the feature and style reconstruction loss which was built on extracting feature from VGG-network. However, feed-forward and back-forward networks for each step reached highly computation time, hence Johnson et al. [12] proposed the use of perceptual loss functions for training feed-forward networks. The run-time comparison showed it achieved 1000x faster under than the original transfer method under different image sizes.

2.3 DAM with Blockchain

The PDM and DAM are separated in our proposed SPChain under IPFS and permissioned blockchain. Zheng et al. [23] proposed a system that securely shared the continuous-dynamic health data. Blockchain module validated the process of accessing data where cloud storage was adapted to provided an off-chain space. Once users uploaded Advanced Encryption Standard encrypted data and

validated by blockchain, buyers were able to buy the data with PHD coins. Thereafter, key keepers would release the key to decrypt via an authenticated channel. However, data was stored in centralized cloud storage and prone to suffer SPOF. On top of that, detailed workflow explanation including consent agreement as well as experimental analysis are necessary.

Faber et al. [6] then proposed Blockchain-based Personal Data and Identity Management System, namely BPDIMS, which integrated with MyData human-centric approach. Three different blockchain layers were in charge of the condition for data exchanges, privacy and consent definition with SC-driven, and hash of verified data. Chowdhury et al. [4] presented a real-time framework for Personal Data Storage (PDS) using permissioned blockchain as the notarization service which mainly including notarization, Audit Trail, Privacy Protection. Involved roles consisted of PDS, Data Custodian, Context provider and Blockchain Client. Completed diagram for sharing and verification of data was discussed whereas the work was still lacked of more privacy discussion and evaluation based on the assumed scenario.

Later, Truong et al. [20] came up with a completed blockchain-based solution for GDPR-compliant PDM with design of SCs and thorough performance evaluation. Core concept of the work was 3A_ledger and log_ledger. The 3A_ledger, which was responsible for authentication, authorization and access control, recorded the pointer of personal data and consent policy versus different roles to an immutable ledger in Hyperledger Fabric (HLF). When Third-Party (TP) or DPs tried to get consents from DSs, policy would be updated from 3A_ledger then responded with the secret key of encrypted data pointer. The GDPR standardizes most stringent personal data protection policy. The process of processing conform the principle of "Personal Data Processing Principles" defined in Article 5 of the GDPR. Three types of blockchain i.e., public, private, consortium cannot guarantee "Storage Limitation" and "Accuracy" [22]. In terms of the regulation, only [19] abides by all the GDPR principles among blockchain platforms.

3 SPCHAIN DESIGN

This section explains the problem definition and design of SPChain and starts with the system architecture. Firstly, major assumptions, the function of components of the digital system, and an in-depth discussion of safely managing digital artworks. Section 3.3 reveals how to combine smart contracts with models in favor of MD to get rewards and excite CR creativity. Finally, the process of personal data under SP and DS's consent will be described. The major Participants' Assumptions of SPChain are $Creator(CR)$, $Collector(CR)$, $ModelDeveloper(MD)$, $ArtGallery(AG)$, $ThirdParty(TP)$.

Problem Definition: In a fair digital artworks creation and trading platform (or any DAMS), the process of handling assets should be secure and transparent. Also, DSs should be guaranteed to know every variation of the assets. Otherwise, the data of DS may be misused without authorization, or even worse – tampered with and destructed. In addition, in response to the effectiveness of the GDPR, more discussion and implementation must be explored, or violators may be fined or forced to suspend operations. On the other hand, many crypto art-related platforms and machine learning model trading platforms exist. The digital art creation and computer vision are tightly integrated, and the separated platforms will cause the maintenance cost to increase and reach less creativity simultaneously.

In this work, SPChain is proposed to securely protect digital assets by leveraging the transparency and immutability of the blockchain. Furthermore, an Off-Chain Storage, IPFS, was used to achieve GDPR compliance. With regard to the combination of artwork creation and machine learning model, container-based virtualization (i.e., Docker) was introduced so that the issue of heterogeneous architecture was resolved. When containers run, art creators can invoke model API inside SCs directly.

3.1 High-Level System Architecture

Figure 1 shows the High-level System Architecture, consists of four modules: Docker Registry, DAMS, Off-Chain Storage, and HLF Blockchain. The DAMS module is used to interact with other participants' modules. The $ArtGallery(AG)$ as an SP provides end-users different functionalities such as updating artwork, transferring artwork, and uploading models and collection views. The data generated by these interactions will be stored in a tamper-free ledger in HLF and Off-Chain Storage (i.e., IPFS). The SPchain implemented five smart contracts to achieve such as confidentiality, integrity, and regulation in DAMS described as follows.

Artwork_CC: Defines Create, Read, Update, Delete (CRUD) of artworks.

Wallet_CC: Manages wallets that store a custom virtual token that is used to represent the medium of exchange.

Model_CC : Stores the address of models in the Docker Registry and offers interface of invoking model API.

3A_CC : 3A_CC is used to manage authentication, authorization, and access-control when different participants want to access the data [20]. Specifically, 3A_CC lets data subjects define their consents.

Log_CC: Log_CC logs the operation toward the data and acts as a second layer of protection to prevent illegal roles from accessing personal data.

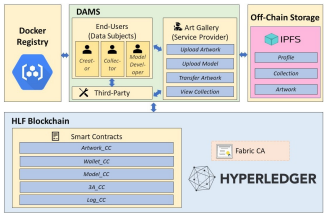


Figure 1: High-level System Architecture of SPChain.

3.2 Design of Managing Artworks

Art management is divided into three phases: initialization phase, uploading or registering phase, and transferring of artwork phase. The initialization phase is used to obtain the consent of personal data from end-user functionalities. Technically off-chain data are stored in the distributed database based on IPFC. $Creator$ and $Collector$ initially have their database to store their collections or creations.

Initial Phase: ***ArtGallery*** Provide the interaction between participants and end-user functionalities. At the moment, ***AG*** plays the role of the data controller or data processor. The sequence of representation shown in Figure 2, ***AG*** first attaches its signature ***Collector*** to end-users for consent request. As soon as the consent is obtained, ***AG*** receives the returned end-users' signature used to encrypt and decrypt the address of orbitdb(step 1-2). Afterwards, ***AG*** updates the policy value in 3A_ledger, which ensures that the end-users' consent is written into the ledger(steps 3-4). The policy and respective permission accessing are defined using the reference key to decrypt the ciphertext(step5). And finally, respond and add the writer in steps 6 and 7.

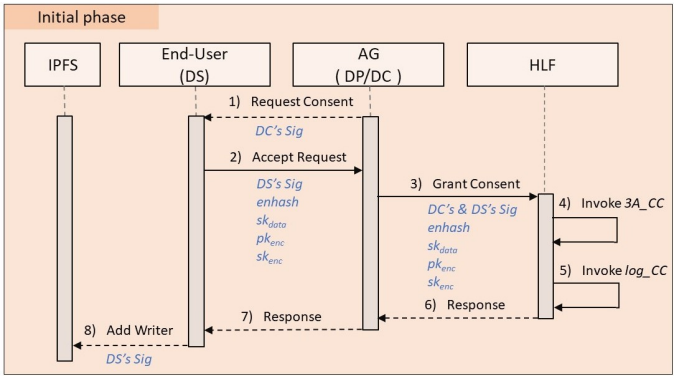


Figure 2: Initial Phase of Granting Consent for AG

Uploading artworks: The process of uploading artworks presented in Figure 3. The artwork includes the metadata, and orbitdb has been established by ***CR*** using Initial Phase. Initially, the artwork is added, multi-hash adding entry is returned. Immediately artwork_CC record proof of provenance established the artwork.

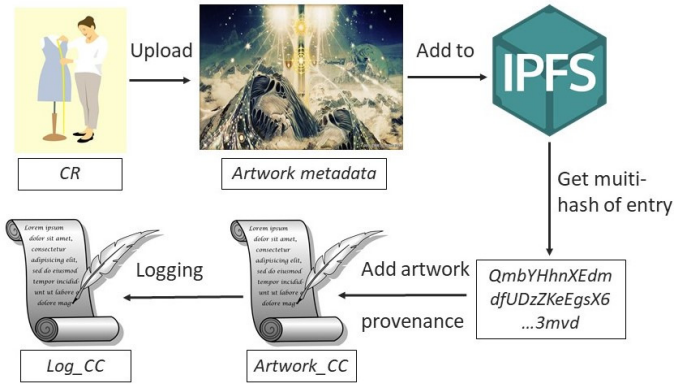


Figure 3: Flow of Uploading Artworks

Transferring artworks: The flow of transferring artwork is represented in Figure 4. In Figure 4, ***AG*** will first look up ***CL*** and ***CR*** of the transferring artwork from the ledger to currency flow. After that, selling the artwork price from orbitdb and passing its roles in monetary flow to wallet_CC.

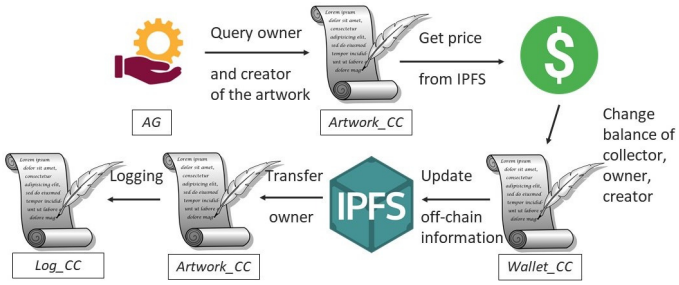


Figure 4: Flow of Transferring Artworks

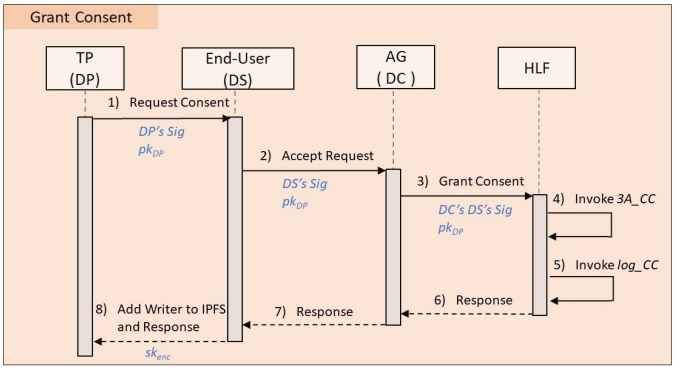


Figure 5: Process of Granting Consent from DS

3.3 Design of Combining Models

The design of combining models can be separated into two parts: uploading the models and invoking the models requested from ***CR***. In order to be felicitous toward more micro-service-oriented architectures nowadays, virtualized container technology is regarded as one of the unstoppable choices since these containers will be independent of each other and make the entire deployment process lighter and easier. The two main parts are used to upload the design of combining models, such as uploading and invoking models. The uploading model is used to upload the source address of the model using Docker Registry, name, developer and associated description of the recorded model.

The invoked model recorded the relevant information of the model into the ledger and started to add flavor to the artwork. At this time, relevant information on models has been listed for ***CR*** to choose from. Taking the GDPR into data portability, three processes are involved: Request Consent, Request Data, and Revoke Consent. The general process is similar to which ***AG***.got the Consent from DS. The method of requesting data from DS is shown in Figure 5. In this process, the request consents are used to obtain the data from off-chain storage, the requested data is used to authenticate and validate, and the revoked Consent is used to access the permission of the DP

4 EXPERIMENTS

In this section the experimental environment settings and reference indicators such as read throughput, write throughput, read latency,

and write latency are described first, and then the experimental results of different functions in the SCs based on the reference indicators will be elaborated. The differences with the previous works are sorted out and the compliance with the GDPR is reviewed, as well as the discussion of the benefits from AI model combination. The experimental environment of blockchain network are Intel i7 processor, 16 GB RAM, Ubuntu 16.04 and Golang programming language for implementing in artwork_CC, wallet_CC, model_CC, 3A_CC, log_CC.

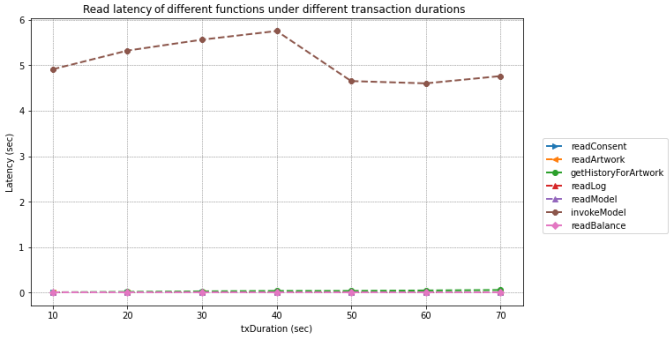


Figure 6: Read Latency of Different Functions under Default Settings

4.1 Experimental Results

We implement each smart contract's function's read latency, write latency, read throughput, and write throughput. Each round's implementation and testing make use of the many functions and related descriptions. For instance, the readArtwork, uploadArtwork, and transfer Artwork functions in Artwork SC are used to implement reading and writing activities. These can be divided into two main categories: querying the ledger (R) and updating the ledger (W). It is noteworthy that invokeModel is evaluated using a rapid style transfer model from [1], whose architecture was suggested by [12]. The model receives a 664 KB input image, which is then converted to a mosaic image. In Hyperledger Fabric, the default configuration for BatchTimeout (BT), which denotes the amount of time to wait for a subsequent transaction to arrive before slicing into network blocks, is set (i.e., 2 seconds). In the meantime, Hyperledger Fabric's MaxMessageCount (MMC), which specifies the amount of transactions that can fit in a single block, is set to the default configuration (i.e., 10 transactions per block). For each of the functions listed above, we extended the transaction sending time (txDuration) from 10 to 70 seconds while monitoring the latency and TPS. Figures 6and Figure 7, illustrate the read and write latencies, respectively. With the exception of invokeModel whose latency spans from 4 to 6 seconds, it can be seen that each function's read latency is almost equivalent to 0 seconds regardless of how long a transaction request takes. The model's inference takes up the majority of this time. In terms of write latency, txDuration shows a modest increase in time. Finally, when txDuration is 70s, the latency for each function varies between 3 and 5 seconds.

The results of the read and write throughputs at various maximum message count (MMC) values are depicted in Figure 8 and

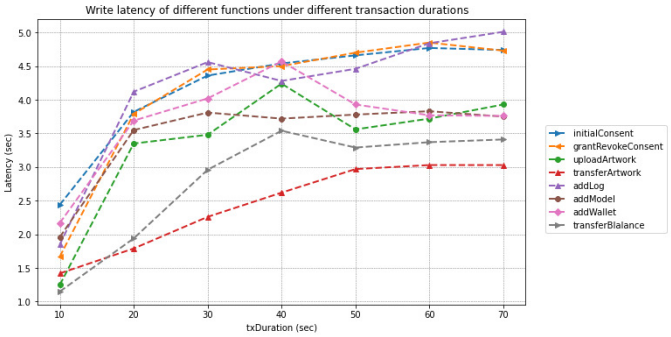


Figure 7: Write Latency of Different Functions under Default Settings

Figure 9, respectively. Except for the rise from roughly 280 TPS to 400 TPS in the case of the function getHistoryForArtwork. we can see that there is no variation in the read throughput compared to the default option (i.e., MMC=10) while increasing MMC to 20. When MMC=18, there is an average improvement of 2 to 3 TPS write throughput. We will use the MMC value of 18, which results in a decreased latency and higher throughput. the part after this, BatchTimeout will be changed to improve performance.

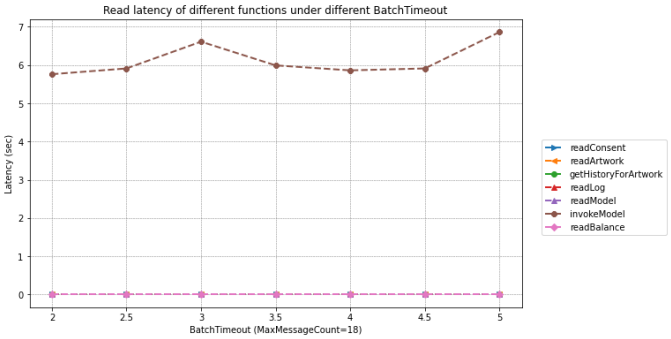


Figure 8: Read Latency of Different Functions under Different BatchTimeout

4.2 GDPR Compliance and benefits of Combining Models

This section discussed the GDPR complaints and the benefits of combing AI models. Combining the AI models benefits the six principles and three rights in the GDPR. The six principles (i.e., Accuracy, Storage limitation, integrity, confidentiality, right to rectification, and right to erasure) and three rights (access, corrections and scope contents) are discussed to manage the relevant rules. Orbitdb is more fault tolerant than conventional centralized databases, and IPFS prevents "all" data from being deliberately erased. Technically, the nodes that request data from an orbitdb hash address will replicate that data, which, if there is no pin or usage, will be actively cleared by IPFS. Because of this, we encrypted both the orbitdb address and the user information. The actual hash address of orbitdb and the plaintexts are only known by the nodes of certain

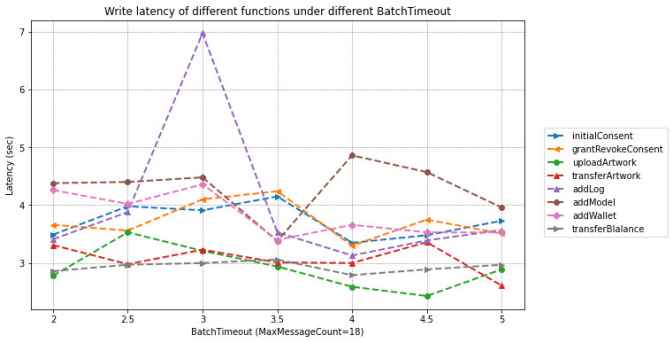


Figure 9: Write Latency of Different Functions under Different BatchTimeout

data subjects, service providers, and third parties (which is more like a private orbitdb).

To increase the inventiveness of the artwork produced, ***CR***can use the models made by ***CR***, as explained in Section 3.3. In most realistic scenarios, the design at this point is frequently not a single process. For instance, ***CR***might acquire the artwork through a producing model before subjecting it to additional processing (e.g., noise removal, style transfer). These processes can run in a distinct container systematically without interfering with one other and communicate with one another until a desirable product is produced.

5 CONCLUSION

In this paper, SPChain was suggested. The three key contributions of SPChain are as follows: (1) A comprehensive design for securely and privately managing the digital assets, where we employ digital artwork as a complete illustration. (3)The fusion of digital assets and AI models using virtualized container technology. (2) The adoption of decentralized Off-Chain Storage (i.e., IPFS) to replace conventional centralized storage to avoid the single point of failure issue. According to the testing results, when transaction time was 50s, MaxMessageCount was 18, and BatchTimeout was 4.5s, the write throughput increased to between 10 and 15 TPS following the MaxMessageCount and BatchTimeout changes. The potential further direction of the research is the blockchain network's architecture configuration needs to be modified to provide for increased stakeholder or organization participation. The suggested SPChain can be used to handle various digital assets, such as pictures, audio files, private information, etc.

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